JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B7, PAGES 3587-3596, JULY 10, 1980

PALEOMAGNETIC RESULTS FROM SOME SOUTHERN AFRICAN KIMBERLITES, AND THEIR TECTONIC SIGNIFICANCE

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Abstract. Paleomagnetic data are reported from kimberlite occurrences of three distinct age groups: (1) National (one pipe), D = 221°, I = -51°, k = 41, α_{95} = 7.2°, age ∿1140 m.y., (2) Swartruggens (three fissures), mean D = 293°, I = -46°, k = 22, α_{95} = 26,9°, age = ∿150 m.y., and (3) six bodies in and adjacent to Lesotho, D = 350°, I = -69°, K = 66, α_{95} = 8.3° with age spread of ∿90 + 5 m.y. Analysis of these and previously published data suggests that most individual kimberlite bodies give acceptable paleomagnetic (as distinct from virtual geomagnetic) poles. This leads to a revised apparent polar wander curve for Africa in upper Mesozoic time and suggests that the time of kimberlite emplacement does coincide with intervals of rapid plate motion, although this cannot be the primary cause of their intrusion.

Introduction

McFadden [1975, 1977] and McFadden and Jones [1977] have reported the results of detailed paleomagnetic study of five kimberlite pipes at and around Kimberley, South Africa. They concluded (1) that the stable remanent magnetization in the kimberlites was primary and acquired penecontemporaneously with emplacement of the pipes and (2) that the emplacement temperature of the kimberlites was generally so low (<350°C), and their cooling history inevitably so rapid, that there was insufficient time to average out secular variation. In light of this the almost identical remanent vector obtained from three Kimberley pipes, De Beers, DuToitspan, and Wesselton, implied that they were emplaced simultaneously.

This paper reports new data obtained from other kimberlites in South Africa and Lesotho which are consistent with the findings of McFadden and Jones. However comparison of our results from Jurassic (∿ 150 m.y.) and Cretaceous (∿90 m.y.) kimberlites with other recently published African paleomagnetic poles of equivalent age supports the interpretation that most kimberlites give true paleomagnetic poles rather than virtual geomagnetic poles. The possible significance of the deduced late Mesozoic apparent polar wander history with respect to the origin of kimberlits will be discussed.

Sampling and Measurement

The location of the kimberlite bodies for which new paleomagnetic data are reported here, together with those previously published, are shown in Figure 1. In this study, samples were either drilled in the field (National, Monastery, and Marakabei) and oriented by sun compass or were hand samples oriented with carpenter's level and Brunton compass. When using the Brunton, every effort was made to avoid local magnetic anomalies either by repeating azimuth measurements from different positions or by reference to topographic maps or the surveyed maps of underground workings (Swartruggens, Letseng). This method of orientation was never ideal, however, and the estimated accuracy of orientation of individual cores is probably no better than 5°. The consistency of the results, notwithstanding these inaccuracies, is testimony to the quality of kimberlite as material for paleomagnetic studies. The data reported in this study were obtained by measurement on a Princeton Applied Research SM1 magnetometer, with ac demagnetization on standard apparatus; no thermal demagnetization has been attempted.

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Magnetic Results

On the basis of the grouping of the natural remanent magnetization (NRM) vectors and the available radiometric age data the paleomagnetic data presented in Table 1 and Figure 2 are discussed in three separate groups.

National Pipe

This pipe (see Figure 1), only 70 m in diameter, has given a mica-isochron age of 1140 ± 16 m.y. (H. L. Allsopp et al., manuscript in preparation, 1979, see Table 2). (All Rb-Sr ages reported have been recalculated, where necessary, using $\lambda = 1.42 \times 10^{-11}$ yr^{-1}). It is located 11 km from the Premier Mine pipe, which was studied paleomagnetically by Jones [1968] and for which an age of 1223 m.y. was reported by Barrett and Allsopp [1973]. Further work (H. L. Allsopp et al., manuscript in preparation, 1979) suggests

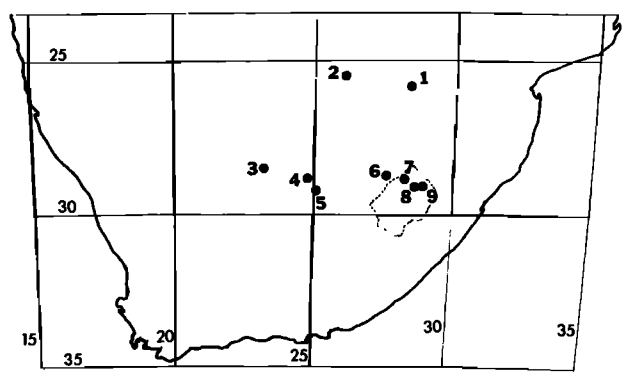


Fig. 1. Map showing location of kimberlite bodies sampled in the study and discussed in text: 1, Premier, National; 2, Swartruggens; 3, Finsch; 4, DeBeers, DuToitspan, Wesselton; 5, Koffiefontein; 6, Monastery; 7, Lipelaneng, Marakabei; 8, Kao, Pipe 100; 9, Letseng 1a Terai; Lesotho is outlined.

Paper number 80B0314.
0148-0227/80/080B-0314$01.00

TABLE 1. Summary of Kimberlite Paleomagnetic Data

Kimberlite	Location	n	NRM D	NRM I	NRM k	NRM J_n	χ	MDF	ODF	D	I	k	α_{95}	South Pole
National	-25.7°S, 28.5°E	11	225	-59	9	7×10^{-3}	1.3×10^{-3}	250	300	221	-51	41	7	-21°S, 245°E
Premier*	-25.7°S, 28.5°E	16	187	-26	30	1.3×10^{-3}			300	186	-24	26	7	-51°S, 217°E
Swartruggens								mean pole n = 2				10	95	-37°S, 234°E
Main	-25.5°S, 26.2°E	6	296	-50	41	2×10^{-5}		350	300	296	-49	56	9	-34°S, 96°E
Changehouse	-25.5°S, 26.2°E	1	292	-60	...	6.1×10^{-6}	3.4×10^{-5}	400	300	289	-62	...	...	-31°S, 79°E
Male	-25.5°S, 26.2°E	2	115	+26	1670	1.1×10^{-5}		350	300	113	+28	1099	8	-27°S, 112°E
Finsch†	-28.4°S, 23.5°E	7							200	313	-47	116	6	-49°S, 101°E
								mean pole n = 4				30	17	-36°S, 97°E
Kao	-29.0°S, 28.6°E	4	356	-7	84	15.2×10^{-3}	2.5×10^{-3}	250	300	359	-74	95	10	-60°S, 30°E
Letseng la Terai	-29.0°S, 28.9°E	8	323	-50	7	7.2×10^{-3}	1.4×10^{-3}	250	300	345	-54	82	6	-76°S, 90°E
Lipelaneng	-28.8°S, 28.3°E	6	348	-60	157	4.8×10^{-3}	3.3×10^{-3}	75	300	345	-61	30	4	-72°S, 66°E
Marakabei	-28.7°S, 27.9°E	7	334	-76	72	5.2×10^{-3}	2.5×10^{-3}	125	300	347	-75	18	14	-55°S, 38°E
Monastery	-28.9°S, 27.5°E	10	5	-77	78	25.3×10^{-3}	3.7×10^{-3}	125	300	21	-74	34	8	-56°S, 8°E
Pipe 200	-29.0°S, 28.6°E	6	342	-70	20	2.2×10^{-3}	2.5×10^{-3}	75	300	336	-71	23	14	-58°S, 55°E
DeBeers†	-28.5°S, 24.8°E	18												-56°S, 50°E
DuToitspan†	-28.5°S, 24.8°E	17												-60°S, 48°E
Wesselton†	-28.5°S, 24.8°E	8												-56°S, 48°E
Koffiefontein†	-29.5°S, 25.1°E	7												-49°S, 40°E
								mean pole n = 10				44	7	-61°S, 44°E

Explanation: n is number of samples; D Declination; I Inclination; K Fisher's dispersion parameter; α_{95} semiangle of cone of 95% confidence (degrees); J_n intensity of NRM (emu/cm^3); χ susceptibility (emu/cm^3 Oe); MDF median demagnetizing field (Oe) that alternating field required to reduce J_n to 50%; ODF Optimal demagnetizing field (Oe) that demagnetizing field at which the best grouping of site vectors was obtained; mean south poles are given, with appropriate statistical parameters.

*Jones [1968].

†McFadden and Jones [1977].

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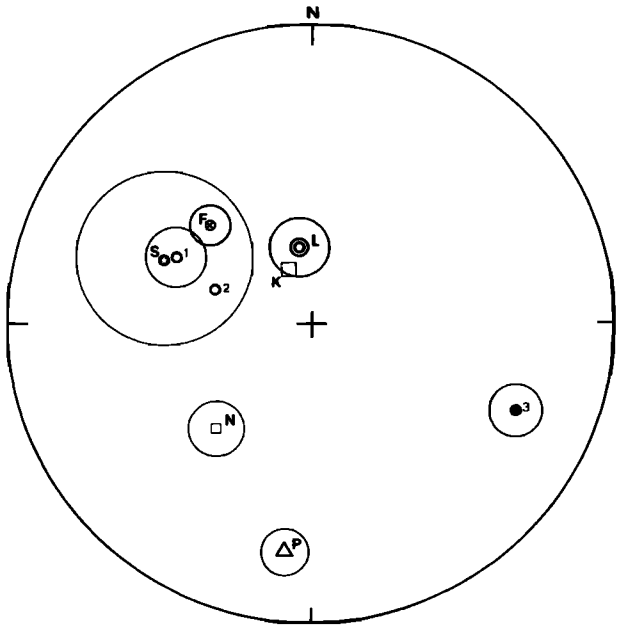


Fig. 2. Equal-area projection summarizing paleomagnetic vectors reported in this study and discussed in text. Open (solid) symbols on upper (lower) hemisphere. L is mean of six Monastery-Lesotho kimberlites reported here; K is mean of DeBeers, DuToitspan, Wesselton, and Koffiefontein, reported by McFadden and Jones [1977]; 1, 2, and 3 are Swartruggens fissure veins, Main, Changehouse, and Male; F is Finsch, reported by McFadden and Jones [1977]. S (star) is mean of three Swartruggens fissures (3 is reversed) plus Finsch; N is National; P is Premier, reported by Jones [1968].

that 1170 m.y. is the best estimate of the time of intrusion of the Premier pipe.

The mean vector obtained from National (D = 221°, I = -51°) is close to but significantly different from the Premier Mine vector (D = 186°, I = -24°). (At Premier, there are several distinct mappable kimberlite units, but whether there is much difference in age is doubtful, H. L. Allsopp, et al., manuscript in preparation, [1979]). Given their close proximity and similarity in age, a close genetic relationship seems likely. If secular variation is averaged out in the magnetization of kimberlites, however (see later section), then the National pipe provides another dated Precambrian paleomagnetic pole for southern Africa; alternatively, it would seem advisable to average the Premier and National results, for an 1150 m.y. pole.

Ito et al. [1978] have reported paleomagnetic results obtained on kimberlites from several pipes, including National. While their data are compatible with previously published data and with those reported here, they were obtained from study of at most two separate samples at each site (only one from National) and hence cannot be regarded as statistically significant.

Swartruggens

Three distinct, parallel, vertical kimberlite fissure veins from 12 to 60 cm wide are exposed in the workings of the Helam diamond mine at Swartruggens in the western Transvaal (Figure 1). The three veins, named Main, Changehouse, and Male, are closely associated (always within 5m) and are locally in direct contact with one another; elsewhere they are separated by relatively thin septa of country rock. Samples from Swartruggens have been dated at 144 ± m.y. by Allsopp and Barrett [1975] and 147 ± 3 m.y. by Allsopp and Kramers [1977]; H. L. Allsopp et al. (manuscript in preparation, 1979) prefer an age of 155.6 ± 9.4 m.y.

Six samples were collected from the Main fissure vein, which is the principal source of ore and is preferentially exposed in the workings. Presuming that they were all penecontemporaneous, no particular effort was made to sample the other fissures, and unfortunately, only two samples were collected from the Male, and one from the Changehouse. These also gave excellent paleomagnetic results (Table 1), with the two samples from the Male fissure being roughly reversed with respect to the others. Assuming field reversal to take at least 1000-20,000 years [McElhinny, 1973, p. 137], this clearly gives an indication of the minimum time interval separating the emplacement of these fissures.

Monastery and The Lesotho Pipes

1. The artificial exposures of the Monastery pipe, in the eastern Orange Free state province of South Africa (Figure 1) with its spectacular display of xenoliths and xenocrysts [Whitelock, 1973], afforded the opportunity of detailed sampling by drilling. The pipe is elliptical in plan, roughly 150 x 50m. While the majority of the cores drilled were kimberlite, three were deliberately drilled (or at least collared) in (1) a garnet lherzolite xenolith, (2) a garnet megacryst, and (3) a xenocryst composed of an intergrowth of ilmenite and diopside. Both of the xenocryst cores measured included some kimberlite matrix, and their NRM is indistinguishable from the other kimberlites. The garnet lherzolite core, however, was homogeneous, and although it was much weaker, it gave the same general vector (Table 1). Allsopp and Barrett [1975] report a phlogopite mica age of 88 ± 4 m.y. for Monastery; Davis [1977] reported a 90.4-m.y. age from zircons.

2. The bulk of the Lesotho samples (Figure 1) were oriented hand samples collected during the Lesotho field trip in connection with the International Conference on Kimberlites in South Africa in 1973. For detailed geological descriptions, see the guidebook and Nixon [1973]. Cores were drilled in two separate exposures of the 2-m-wide Marakabei dike during a subsequent visit in May 1974.

Samples were collected from a natural outcrop at Lipelaneng, from underground workings at Letseng la Terai (pipe, 450 x 300 m), in surface workings at Kao (pipe, 400 x 500 m), and pits at Pipe 200 (150 x 100 m). Samples from Letseng included basaltic xenoliths, basalt at the contact, and two generations of kimberlite. The Letseng kimberlite samples, in particular, proved very fragile, expanding considerably on exposure to water during the drilling in the laboratory. Nevertheless, oriented cores from five kimberlite samples, in addition to three basalt xenoliths, were obtained. The Kao samples were all from xenoliths of Stormberg (Karroo) basalt well within

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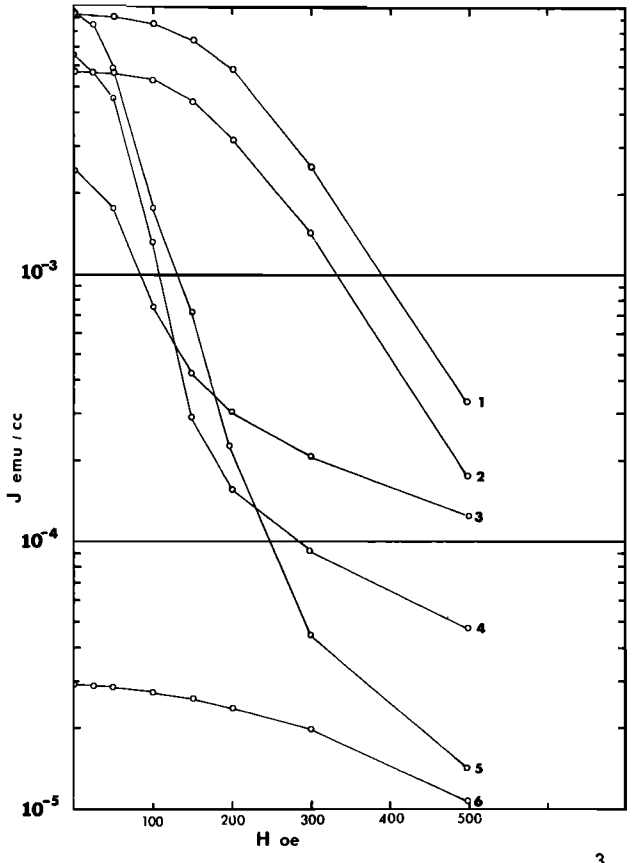


Fig. 3. Intensity of NRM remaining (emu/cm^3) as a function of demagnetizing field intensity (H, Oe): 1, Letseng la Terai, sample 2; 2, National, sample 2; 3, Pipe 100, sample 1; 4, Monastery, sample 2; 5, Marakabei, sample 4; 6, Swartruggens, sample 3, Main seam.

the kimberlite body; the single sample from the immediate wall rocks gave anomalous results and was omitted. Pipe 200 samples were all hard basaltic kimberlite from exposures in pits. Lipelaneng is a natural exposure of a xenolith-charged, 13-m-wide dikelike body with abundant dunitic nodules and some crustal xenoliths cemented by a rather basaltic matrix.

Both within- and between-site dispersion (Table 1) is very low, and the mean vector from these six pipes is not significantly different from the mean of the four Kimberley pipes (DeBeers, Du Toitspan, Wesselton, and Koffiefontein) reported by McFadden and Jones [1977].

In view of the petrologic diversity and wide distribution (over 150 km) of the Lesotho kimberlite bodies studied, it seems unlikely that they were all emplaced exactly simultaneously, yet the close grouping of poles suggests that they are all relatively contemporaneous. Their mean pole therefore is probably a representative paleomagnetic pole, ∿90 m.y.B.P. The paleomagnetic significance of the individual poles is discussed in a later section.

Magnetic Properties of Kimberlite

As demonstrated by the data in Table 1 and in previous studies [Jones, 1968; McFadden, 1977; McFadden and Jones, 1977], kimberlite and xenoliths in kimberlite (both crustal and mantle) give consistent and relatively stable paleomagnetic results. Knowledge of the carrier mineral (or minerals) of this NRM, and its origin, is crucial to the complete understanding of its paleomagnetic significance.

The average intensity of NRM, median demagnetizing field (MDF), and magnetic susceptibility of the kimberlites are given in Table 1, and the alternating field (AF) demagnetization curves of representative specimens are shown in Figure 3. The MDF varies from about 75 to 400 G, but even in those samples in which it is lowest, (Monastery, Marakabei), the direction of the remanent vector showed little change until above 300 Oe, and the direction of the NRM's of all kimberlites is very similar to the stable, demagnetized orientation (Table 1). The present earth's field direction in Lesotho (D = 340°, I = 65°) is close to the stable direction of the Monastery-Lesotho kimberlites [see McFadden and Jones, 1977], so any viscous component which may be present in the less stable samples from this group (Monastery, Marakabei) could not be distinguished. Those kimberlites with significantly different primary remanence direction (Swartruggens and National) are extremely stable (Figure 3), and there is probably very little viscous magnetization present.

Saturation magnetization (J/H) and its temperature dependence (J_s/T) curves have been run on several kimberlite samples (Figure 4). In each, an easily saturatable component is present (up to 1.7 emu/g, presumably a ferrimagnetic spinel), the specific amount of which correlates roughly with the intensity of NRM (J_n) and susceptibility of the samples (Table 1).

Curie temperatures, measured in air in a 3500-G field, range from about 420°C in a Monastery sample to 590°C in a Letseng la Terai kimberlite. This would conventionally be attributed to varying amounts of ulvospinel in solid solution in magnetite. In polished section, the Letseng sample was found to contain abundant primary titaniferous magnetite (in addition to a variety of other opaque phases) which had undergone intense subsolidus oxidation, with dense, fine ilmenite lamellae and development of what appears to be maghemite (this texture may well account for the stability of this sample). The high Curie point and the fact that the room temperature moment decreased by about 20% after heat treatment to 700°C (suggesting oxidation of magnetite or inversion of maghemite), is consistent with these observations.

Other kimberlite polished thin sections studied revealed a bewildering variety of primary and secondary rhombohedral and cubic opaque phases, individually zoned, or one occurring as a reaction rim on the other.

Haggerty [1975] has given a detailed report on the mineralogy and chemistry of an extensive suite of kimberlite samples (including pipes sampled in this study). He emphasizes the chemical and textural complexity of the secondary spinels and the compositional zoning which they almost invariably display. In addition to spinels occurring as xenocrysts and as primary groundmass phases, he distinguishes the following secondary varieties [Haggerty, 1975, p. 296]: 'secondary spinel mantles on xenocrysts and groundmass spinels; secondary

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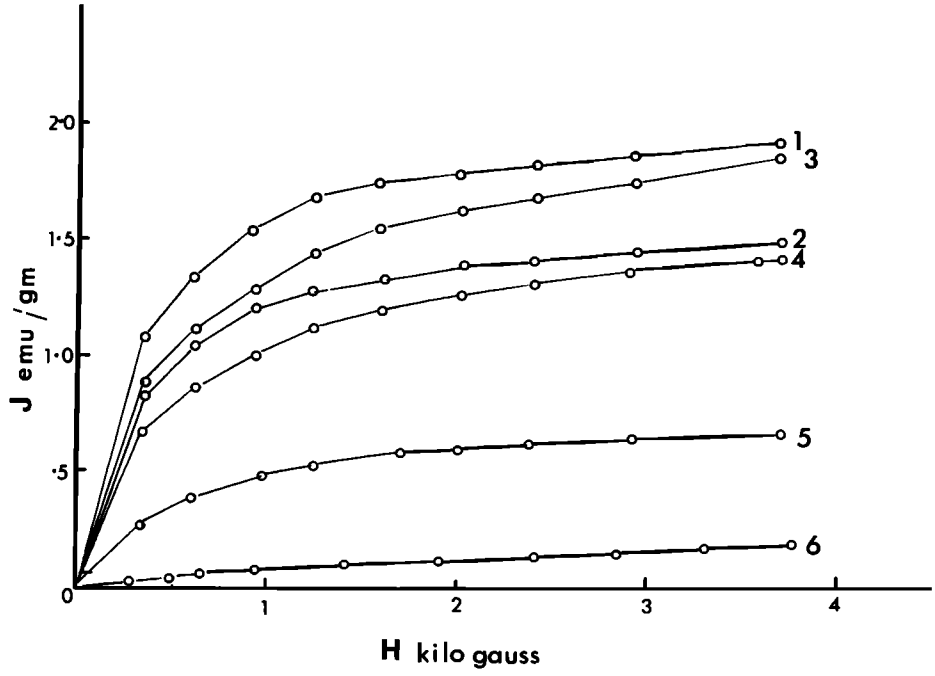


Fig. 4a. Magnetic moment J (emu/g) as a function of aplied field H (gauss); numbering: 1, Monastery, sample 1; 2, Monastery, sample 9; 3, Marakabei, sample 6; 4, National, sample 2; 5, Letseng, sample 2; 6, Swartruggens, sample 5 (Main seam).

spinels in association with perovskite in picro-ilmenite reaction mantles; secondary spinels in garnet rims and in partially serpentinized olivine; and exsolved spinels in picro-ilmenite.' Although the bulk of the spinels are in the compositional field of $MgCr_2O_4$-$FeCr_2O_4$-$MgAl_2O_4$-$FeAl_2O_4$ (and are presumably nonmagnetic), he reports that the zoning trend is always toward a late-stage enrichment in Ti and Fe (as Fe^{3+}). Thus it would appear that magnetic spinels are always the last to form, whether by exsolution, solid-liquid, or solid-gas reaction. That the Curie temperatures we measure should range from that of an ulvospinel-bearing magnetite (Usp ~ 20%) to relatively pure magnetite (or maghemite) is also implicit in the mineralogic diversity that

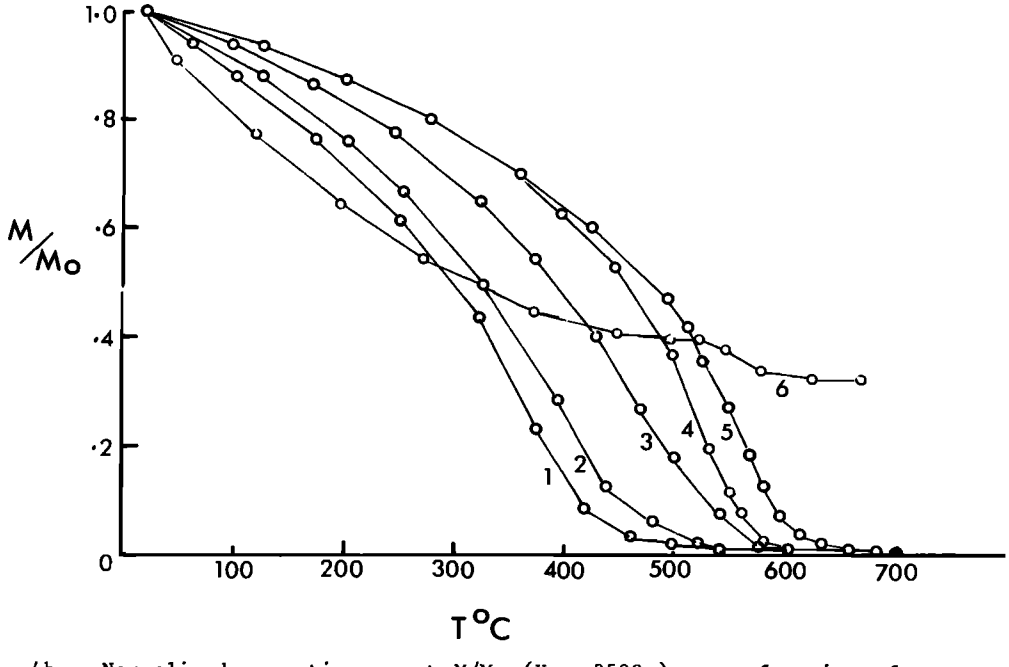


Fig. 4b. Normalized magnetic moment M/M_o (H = 3500g) as a function of temperature. Numbering as for Figure 4a. Note spectrum in Curie temperatures; the Swartruggens sample contains very little magnetic mineral, and its J_s/T curve is dominated by paramagnetism of the other minerals.

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Haggerty [1975] reports and in the complex physiochemical reactions which kimberlites evidently experience during and after their emplacement. If the magnetic minerals form at this time, the remanent magnetization is, at least in part, thermochemical in origin. The duration of this period of secondary spinel formation, postemplacement, dictates the duration of the magnetization intervals. This aspect is considered in the next section.

Acquisition of Magnetism

McFadden [1977] has concluded that the temperature of emplacement of the kimberlites he studied was less than $\sim 350^{\circ}C$. In addition, knowing the dimensions of the pipe and assuming cooling by conduction only, McFadden and Jones [1977] calculate that the DeBeers pipe, for example, would cool to ambient temperatures in less than 500 years. This, they point out, would be insufficient time to average out secular variation if all the magnetization was acquired during this interval. They also emphasize the low dispersion of specimen magnetization vectors at individual pipes, which is less than that expected if each specimen represents one magnetization instant and the sample collection is temporally distributed over a complete secular variation cycle. They interpret the low dispersions as evidence that as a rule, in the sampling of individual kimberlites, secular variation is not averaged out. The near identity of the vector they report from the Bultfontein, DeBeers, and Wesselton pipes is therefore considered to reflect their simultaneous emplacement. Alternatively, however, the low dispersion could be considered to reflect the fact that even at the individual sample level, the magnetization interval does approach that required to average out secular variation (10^4 years, McElhinny [1973]).

We favor this view, and the evidence considered to support it is summarized below.

1. The three pipes shown by McFadden and Jones [1977] to have essentially identical NRM's are reported to be 92.0 (DeBeers), 83.8 (DuToitspan) and 90.3 (Wesselton) m.y. by Davis [1977, 1978]. The age reported by Allsopp and Barrett for a DeBeers-Bulfontein-DuToitspan composite is 88 $\pm$ 3 m.y., whereas they report Wesselton as 82 $\pm$ 3 m.y. Although these pipes could be exactly contemporaneous within the error margins of the radiometric techniques, the limits of these errors are strained. We consider it more likely that these pipes are singificantly different in age and therefore the similarity in the NRM's implies that the secular variation source of vector divergence has been averaged out.

2. In their emplacement temperature and cooling models, McFadden and Jones [1977] assume that each kimberlite was emplaced in one single intrusive pulse (and not as a succession of intrusions), and they acknowledge that multiple intrusions would at least prolong the cooling interval. It should be noted that multiple intrusions are commonly observed in kimberlites (e.g., Monastery, Premier, Letseng), and thus in some kimberlites, at least, the duration of the cooling interval may be considerably longer than the simplified cooling models would suggest.

As disussed previously, however, the NRM of kimberlite appears to be carried by secondary magnetic spinels produced by various subsolidus reactions affecting oxide and silicate phases after emplacement. The NRM is therefore thermochemical in origin, and the period over which these subsolidus reactions occur is more relevant to estimates of the magnetization interval, rather than purely thermal models.

For a long time after their initial emplacement, kimberlite pipes are likely to serve as conduits for the continued escape of primary volatiles, and ferromagnetic oxides are a probable byproduct of the interaction of these with ferromagnesian minerals. Furthermore, serpentinization, the most important of these alteration processes, is exothermic; once started, this process tends to be self-sustaining, since the heat release may drive the convective circulation of groundwater.

We have no quantitative estimate of how long these low-temperature alteration - and thermochemical magnetization - processes may continue. We suggest, however, that with multiple intrusions of kimberlite, slow hydrothermal alterations, the magnetization interval in larger kimberlites may be very much longer than the simple cooling (TRM) models would suggest. For this reason we claim that in most kimberlite bodies, secular variation is likely to be averaged out.

3. McFadden and Jones [1977] report the presence of a reversed component of magnetization in both kimberlite and accidental inclusion samples from the Koffiefontein pipe. They found no experimental evidence to support a suggestion of partial self-reversal as the origin of this component. As they inclined to the view that the duration of the magnetization interval in these kimberlites was short, they had no explanation for this reversed component. If this reversed component is due to field reversal, however, then the magnetization interval must have been sufficiently long (10,000 years?) to permit this to occur in each individual Koffiefontein sample - more than sufficient to average out the nondipole component of secular variation.

4. As may be seen in Table 1 and Figure 2, the mean of the six Lesotho (plus Monastery) kimberlites sampled in this study ($D = 350^{\circ}$, $I = 69^{\circ}$, $k = 66$, $\alpha_{95} = 8^{\circ}$.) is not significantly different from those at Kimberley, and their collective circular standard deviation ($81/k^{\frac{1}{2}}$) is also very low (10.0). Such close grouping of remanent vectors from igneous bodies of such a range in size and petrological type, distributed over such a large area (from Kimberley to Lesotho, 300 km) is to us more consistent with the hypothesis that secular variation has been more or less averaged out in each rather than that they were all emplaced almost simultaneously. The scatter may include some secular variation component but could as well be due to experimental error and/or apparent polar wander.

5. Our results from the three Swartruggens fissures, on the other hand, show two to be normal and one reversed; the Changehouse and Male fissures, however, are represented by only one and two samples, respectively, and these individual mean vectors cannot be considered significant. Nevertheless, the two Male samples are in good agreement, and we consider the reversal to be real, but their mean vector is clearly not antiparallel to that of the normal fissures. Either secular variation has not been averaged out in the

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TABLE 2. Kimberlite Radiometric Age Data

Kimberlite Pipe	Age, m.y. Mica Rb-Sr*	Age, m.y. Zircon U-Pb	References†
Premier			
Type 1	1223		1,3
Type 2	1174		3
Best estimate overall	1170		4
National	1155 ± 30		3
	1140 ± 16		4
Swartruggens	144 ± 4		2
	147 ± 3		3
	155.6 ± 9.4		4
Roberts Victor	124 ± 3	92.2	2, 6
Finsch	122, 129, 162	94.1	4, 5
DeBeers	88 ± 3	92.0	2, 5, 6
Bultfontein	88 ± 3	91.2	2, 5, 6
DuToitspan	88 ± 3	83.8	2, 7
Wesselton	82 ± 3	90.3	2, 5, 6, 7
Monastery	88 ± 4	90.4	2, 5, 6
Koffiefontein		90.4	7

*Rb-Sr = 1.42 x 10^{-11} yr^{-1}.
†References are 1, Barrett and Allsopp [1973]; 2, Allsopp and Barrett [1975]; 3, Allsopp and Kramers [1977]; 4, H. L. Allsopp et al. (manuscript in preparation, 1979); 5, Davis et al. [1976]; 6, Davis [1977]; and 7, Davis [1978].

Swartruggens fissures, contrary to what has been argued above for the Lesotho pipes, or the Male fissure is significantly different in age. Skinner and Scott [1979] have pointed out that the Male fissure contains leucite, sanidine, and nepheline, in addition to much olivine and phlogopite, and is best classified as a lamprophyre rather than a kimberlite (as are the Main and Changehouse fissures). Such a petrologic difference is consistent with the Male fissure being of distinctly different age and hence possibly explaining the lack of antiparallelism. H. L. Allsopp (personal communication, 1979) feels their present radiometric data are insufficient to determine whether or not there is significant difference in age between the Main and Male fissures. On the other hand, the Swartruggens fissures are so very narrow (12-60 cm) that an abbreviated magnetization interval is particularly likely. Indeed, despite their close proximity and serial emplacement the reversal strongly suggests that their NRM was acquired rapidly and penecontemporaneously with the emplacement of each.

We favor this view and conclude that because of their small size, secular variation has not been averaged out in the Swartruggens fissures. While acknowledging the inadequate sampling of the Male and Changehouse fissures, we feel the best ~150-m.y. paleomagnetic pole is obtained by averaging all three (Table 1).

We conclude that the evidence favors the hypothesis that except in very small bodies like the Swartruggens fissures, kimberlite magnetization does tend to average out secular variation and that the pole calculated from the mean vector for each individual pipe does approximate a paleomagnetic pole. We prefer this as the explanation for the relatively small dispersion of roughly contemporaneous kimberlite poles rather than that they were all emplaced almost exactly simultaneously and each recorded only the same partial segment of the secular variation cycle.

Kimberlite Ages

The published radiometric ages for those Southern African kimberlites sampled in this and other paleomagnetic studies are listed in Table 2. The dates were obtained by two methods: Rb-Sr on phlogopitic micas [Allsopp and Barrett, 1975; Allsopp and Kramers, 1977] and U-Pb on zircons [Davis, 1977, 1978]. The ages for those kimberlites which have been dated by both techniques are generally in good agreement, the Kimberley-Lesotho pipes clustering around 90 ± 5 m.y. One particular exception is Roberts Victor (for which, unfortunately, no paleomagnetic data are available), which is reported to be 124 ± 3 m.y. years by Rb-Sr and 92.2 m.y. by U-Pb. The Finsch pipe, with zircons reported as 94.1 m.y., gave mica ages of 122, 129, and 162 m.y. Mixing of equilibrated and unequilibrated mantle micas is considered to be the cause of the older ages (H. L. Allsopp et al., manuscsript in preparation, 1979). However, as discussed below, it is difficult to reconcile the paleomagnetic data from Finsch with the younger zircon age.

The Finsch Problem

McFadden and Jones [1977] report a pole for the Finsch pipe which is significantly displaced from those of the four pipes close to Kimberley. As pointed out above, they believe that secular variation is not averaged in individual kimberlites and attribute this difference to that cause. Their paleomagnetic pole is the mean of all five kimberlites. By our argument above, Finsch

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TABLE 3. Average Jurassic-Cretaceous Paleomagnetic Poles for Africa

Pole	Formation	Age	Pole Position	α_{95}	Reference
1	Stormberg (Karroo)	(154-190)	65°N, 262°E	11.7	McElhinny [1973]*
2	Hoachanas	161-173	62°N, 252°E	7	Gidskehaug et al. [1975]
3	Mateke Hills Complex	~168	59°N, 260°E	8.3	Gough et al. [1964]
4	Central Atlas Intrusives	156-157	53°N, 261°E	24	Hailwood and Mitchell [1971]
5	Moroccan volcanics	Lower Cretaceous	44°N, 251°E	10	Bardon et al. [1973b]
6	Swartruggens and Finsch	~150	36°N, 277°E	17.2	this paper
7	Kaoko lavas	110-128	48°N, 267°E	2	Gidskehaug et al. [1975]
8	Mlanje Massif	116-128	60°N, 262°E	12	Briden [1967]
9	Lupata Volcanics	106-111	62°N, 260°E	3.5	Gough and Opdyke [1963]
10	Wadi Natash	81-90	64°N, 218°E	4.2	El Shazly and Krs [1973]
11	Volcanics, Sicily	71-81	63°N, 229°E	3.8	Schult [1973]
12	Lesotho-Kimberley pipes	~90	61°N, 224°E	7.4	this paper
13	Moroccan sediments	Lower to Upper Cretaceous	75°N, 217°E	5.5	Hailwood [1975]
14	Nubian sandstone	Upper Cretaceous	79°N, 220°E	4	Schult et al. [1978]
15	Red Siltstone	Upper Cretaceous?	79°N, 208°E	6	Gough and Opdyke [1963]

* Average of the lower Jurassic poles (9.1, 9.2, 9.3).

alone provides a valid paleomagnetic pole. We note that the Finsch pole is quite close to the mean pole from the Swartruggens fissures, dated at ~150 m.y., and hence we would assert that the Finsch pipe is older than those at Kimberley. H. L. Allsopp et al. (manuscript in preparation, 1979) obtain discrepant ages of 122, 129, and 162 m.y. from different samples from Finsch. An age of ~125 m.y. would be more consistent with our paleomagnetic correlation of the Finsch pipe with the Swartruggens, rather than the Kimberley activity. Davis [1977, 1978], however, reports a 94.1-m.y. zircon age for Finsch, which should therefore give the Kimberley-Lesotho pole; we have no explanation for this anomaly. Of the 10 closely grouped kimberlite paleomagnetic poles from Kimberley and Lesotho, 5 have been dated by either micas, or zircons, or both (Table 2), and all ages fall within the range 90 ± 5 m.y. Finsch is the only paleomagnetic exception to this, and the radiometric ages by the two methods are inconsistent. Not understanding this, we have chosen to exclude the Finsch pole from the ~90-m.y. average and have included it in the ~150-m.y. average. Its omission from the one and inclusion in the other would not significantly change the mean vectors assumed in subsequent discussion.

Further research toward resolution of such age and paleomagnetism discrepancies call, for example, for zircon ages from the Swartruggens fissures to see if they agree with the mica ages; unfortunately, there appear to be no zircons in those fissures (H. L. Allsopp, personal communication, 1979). Also it would be desirable to obtain paleomagnetic data from the Roberts Victor pipe, for which the two radiometric techniques also give different ages. Unfortunately, at time of writing, this mine is flooded. If it is ever dewatered, the aquisition of samples should be of high priority. Finally, it is hoped that radiometric age determination on Finsch samples, by both methods, will be repeated. At present the discrepancy remains unresolved.

Late Mesozoic Polar Wander Curve for Africa

The poles from the Kimberley and Lesotho kimberlite pipes coincide with others determined from the Wadi Natash volcanics in Egypt [El Shazly and Krs, 1973] and upper Cretaceous volcanics from Sicily (which is considered to be an extension of the North African Craton) [Schult, 1973] (see Table 3 and Figure 5). The age reported for these two volcanic units (90-70 m.y.) overlaps the ages for the kimberlite pipes, and the similarity of their vectors strongly suggests, as argued above, that the kimberlites give true paleomagnetic rather than virtual geomagnetic poles. On the other hand, they are significantly different from other supposedly upper Cretaceous palemagnetic poles [Gough and Opdyke, 1963; Hailwood, 1975; Schult et al., 1978]; as these are much closer to the present geographic pole (Table 3 and Figure 5), they are presumed to be younger in age. The kimberlite poles are also distinctly separated from those obtained from the slightly older Lupata volcanics (106-111 m.y., Gough and Opdyke [1963]) and the Mlanje Massif (116-128 m.y., Briden [1967]). The distribution of all these poles is

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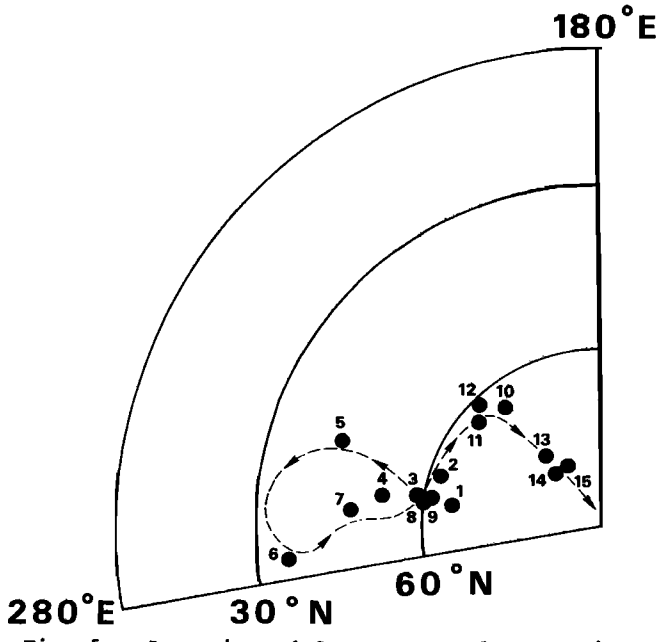


Fig. 5. Jurassic and Cretaceous paleomagnetic poles for Africa and proposed apparent polar wander path. References given in Table 3.

consistent with the occurrence of a hairpin in the African polar wander curve, between 110 and 60 m.y., with the kimberlites poles near the tip of this hairpin (Figure 5).

The paleomagnetic poles from Swartruggens fissures and the Finsch pipe [McFadden and Jones, 1977] form a low-latitude cluster that coincides with those reported for the younger Kaoko lavas (110-128 m.y., Gidskehaug et al [1975]), for the Central Atlas extrusives of Morocco (156-157 m.y., Hailwood and Mitchell [1971]), and for the lower Cretaceous Beni Mellal volcanics of Morocco (Bardon et al. [1973] and Table 3 and Figure 4). These lower-latitude poles are bracketed, on the one hand, by the younger Lupata and Mlanje poles, and on the other, by the older paleomagnetic poles of Mateke Hills (168 m.y., Gough et al [1964]) and of Hoachanas (161-173 m.y., Gidskehaug et al. [1975]). This pattern suggests the occurrence of an earlier loop in the African polar wander curve, beginning around 160 m.y. and ending at roughly 120 m.y. (Figure 4). Although the evidence for this loop is not conclusive, it should be be noted that a similar loop has been proposed for the South American polar wander curve for approximately the same period of time [Schult et al., 1978]. Rotation of South America to its predrift, early Triassic position brings these two loops into approximate coincidence.

The question of whether secular variation has or has not been averaged out in these studies is crucial to their correct interpretation. We have argued the case for the larger kimberlite intrusions above, and the authors of the other pertinent studies likewise feel that their data represent true paleomagnetic poles. The studies of the Egyptian [El Shazly and Krs, 1973] and Sicilian [Schult, 1973] volcanics, for example, represent the mean of nine stratigraphically separated sites and 19 individual flows, respectively. It seems likely that these too are true paleomagnetic poles, and a similar conclusion is valid for most of the other data. However, more detailed studies with close age control will be required for final confirmation.

Kimberlites and Plate Tectonics

With regard to its paleomagnetic record, Africa, until quite recently, was thought to have experienced a 'quasi-static' interval [Briden, 1967] lasting approximately from 200 to 100 m.y. B.P., during which its pole appeared to have wandered very little. The data compiled here and the interpretation given them imply much greater apparent mobility for Africa, at least during the latter part of this interval. Boyd [1973] suggested that changes in plate motion may cause shear heating and kimberlite production in the asthenosphere. It is clear that the Cretaceous kimberlite poles reported here do fall at the apices of the inferred loops or hairpins in the apparent polar wander curve proposed. (The Premier and National poles coincide in time with a period of rapid apparent polar wander in Piper's [1976] Precambrian polar wander curve for Africa.) As such, kimberlite emplacement may indeed coincide with periods of most active plate motion. Yet as Crough et al. [1979] point out, if vigorous plate motion alone is the cause of kimberlites, why are they not ubiquitous? Crough et al. propose that kimberlites are emplaced above hot spots. Reconciliation of all these data would require that vigorous (and perhaps erratic) plate motion might also be caused by hot spot activity, or at least a sudden outburst, in one part of a plate. Such tentative speculation, however, serves only to suggest potentially rewarding future research.

Acknowledgements. Field sampling was performed during 1973-1974, when R. B. H. was a visiting Fellow at the Bernard Price Institute for Geophysical Research, University of the Witwatersrand, South Africa. L. O. Nicolaysen is thanked for his hospitality, support, and encouragement. Magnetic measurements were made in the Rock Magnetism Laboratory at Princeton University (supported by NSF grant EAR76-22644) by Naoma Dorety, whose assistance is much appreciated.

References

Allsopp, H. L., and D. R. Barrett, Rb-Sr age determinations on South African kimberlite pipes, Phys. Chem. Earth, 9, 605-618, 1975.

Allsopp, H. L., and J. D. Kramers, Rb-Sr and U-Pb age determinations on Southern African kimberlite pipes, paper presented at the 2nd International Kimberlite Conference, sponsored by Am. Geophys. U., and others; Santa Fe, New Mexico, 1977.

Bardon, C., A. Bossert, R. Hamzeh, and M. Westphal, Etude paleomagnetique de formation du Trias et du Jurassique du Maroc et du Sahara, C. R. Acad. Sci., Ser. D, 275, 2357, 1973.

Barrett, D. R., and H. L. Allsopp, Rubidium-strontium age determinations on South African kimberlite pipes, paper presented at the 1st International Kimberlite Conference, sponsored by Int. Assoc. Geochem. Cosmochim., and others; Univ. Cape Town, South Africa, 1973.

21562202b, 1980, B7, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB085iB07p03587 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Boyd, F. R., Origin of the lherzolite nodules in the kimberlites of northern Lesotho, paper presented at the 1st International Kimberlite Conference, sponsored by Int. Assoc. Geochem. Cosmochim., and others; Univ. Cape Town, South Africa, 1973.

Briden, J. D., A new palaeomagnetic result from Lower Cretaceous of East-Central Africa, Geophys. J. Roy. Astron. Soc., 12, 375-380, 1967.

Davis, G. L., T. E. Krogh, and A. J. Erlank, The ages of zircons from kimberlites from South Africa, annual report, pp. 821-824, Carnegie Geophys. Lab., Washington, D.C., 1976.

Davis, G. L., The ages and uranium contents of zircons from kimberlites and associated rocks, paper presented at the 2nd International Kimberlite Conference, sponsored by Am. Geophys. U., and others; Santa Fe, New Mexico, 1977.

Davis, G. L., Zircons from the mantle, Open File Rep. 78-701, pp. 86-88, U.S. Geol. Surv., Reston, Va., 1978.

El Shazly, E. M, and M. Krs, Paleogeography and paleomagnetism of the Nubian sandstone, Eastern desert of Egypt. Geol. Rundsch., 62, 212-225, 1973.

Gidskehaug, A., K. M. Creer, and J. G. Mitchell, Paleomagnetism and K-Ar ages of the South-West African basalts and their bearing on the time of initial rifting of the South Atlantic Ocean, Geophys. J. Roy. Astron. Soc., 42, 1-20, 1975.

Gough, D. I., and N. O. Opdyke, The paleomagnetism of the Lupata alkaline volcanics, Geophys. J. Roy. Astron. Soc., 7, 457-468, 1963.

Gough, D. I., A. Brock, D. L. Jones, and N. D. Opdyke, The paleomagnetism of the Ring Complexes at Marangudzi and the Mateke Hills, J. Geophys. Res., 69, 2499-2507, 1964.

Haggerty, S. E., The chemistry and genesis of opaque minerals in kimberlites. Phys. Chem. Earth, 9, 295-308.

Hailwood, E. A., The paleomagnetism of Triassic and Cretaceous rocks from Morocco, Geophys. J. Roy. Astron. Soc., 41, 219-235, 1975.

Hailwood, E. A., and J. G. Mitchell, Palaeomagnetic and radiometric dating results from Jurassic intrusions in south Morocco, Geophys. J. Roy. Astron. Soc., 24, 351, 1971.

Ito, H., K. Tokeida, K. Suwa, and S. Kume, Remanent magnetization of Precambrian and Cretaceous kimberlites in South Africa, Geophys. J. Roy. Astron. Soc., 55, 123-130, 1978.

Jones, D. L., Paleomagnetism of the Premier Mine Kimberlite, J. Geophys. Res., 73, 6937-6944, 1968.

McElhinny, M. W., Paleomagnetism and Plate Tectonics, Cambridge Earth Sci. Ser., 356 pp., Cambridge University Press, New York, 1973.

McFadden, P. L., A paleomagnetic study of Cretaceous kimberlite occurrences in the northern Cape, Ph.D. thesis, Univ. of Rhodesia, Salisbury, 1975.

McFadden, P. L., A palaeomagnetic determination of the emplacement temperature of some South African kimberlites, Geophys. J. Roy. Astron. Soc., 50, 587-604, 1977.

McFadden, P. L., and Jones, D. L., The palaeomagnetism of some upper Cretaceous kimberlite occurrences in South Africa, Earth Planet. Sci. Lett., 34, 125-135, 1977.

Nixon, P. H. (Ed.), Lesotho Kimberlites, Cape and Transvaal Printers, Ltd., Cape Town, 1973.

Schult, A., Palaeomagnetism of Upper Cretaceous volcanic rocks in Sicily, Earth Planet. Sci. Lett., 19, 97-100, 1973.

Schult, A., H. C. Soffel, and A. G. Hussain, Palaeomagnetism of Cretaceous Nubian Sandstone, Egypt, J. Geophys., 44, 333-340, 1978.

Skinner, E. M. N., and B. H. Scott, Petrography, mineralogy and geochemistry of kimberlite and associated lampophyre dykes near Swartruggens, Western Transvaal, R.S.A., paper presented at the Kimberlite Symposium II, Cambridge University, Cambridge, England, 1979.

Whitelock, T. K., The Monastery Mine Kimberlite pipe, in Lesotho Kimberlites, edited by P. H. Nixon, pp. 214-218, Cape and Transvaal Printers Ltd., Cape Town, 1973.

(Received July 16, 1979;
revised January 11, 1980;
February 13, 1980.)

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